BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: KT-XII-21-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** Code C

Note: Q.1 Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. No marks will be awarded for cutting, erasing or over writing.

Qi. If dy/dx = cosx, then y = ...

(A) –sinx+c
(B) –sinx
(C) xcosx+c
(D) sinx+c

Qii. Domain f(x,y) = 1/√(x – y) is.....

(A) x>y
(B) x≥y
(C) y≥x
(D) y>x

Qiii. The function $ax^2+2hxy+by^2$ is homogeneous of degree .....

(A) 1
(B) 2
(C) 0
(D) None of these

Qiv. ∫(from 3 to 2) 5dx = ...

(A) –32
(B) 32
(C) Zero
(D) 5

Qv. If g(R) = $(R^3 – R^2 +6)/(R-3)$ then g(3) = ...

(A) 5
(B) 0
(C) 6
(D) None of these

Qvi. If F(t) = t^2i+j-sintk and G(t) = ti+1/j+5k then (F.G)t = ...

(A) t^3+1-5 Sint
(B) t^3-5sint
(C) [illegible in source scan]
(D) Zero

Qvii. ∫du/9u = ?

(A) lnu+c
(B) lnu+c
(C) 1/9 lnu+c
(D) None of these

Qviii. ∫(10+a)^x dx = ?

(A) (10+a)^x + c
(B) Zero
(C) 10x+c
(D) 10+a

Qix. Slope of a line perpendicular to 6x=y-3 is .....

(A) 6
(B) 1/6
(C) 3
(D) –1/6

Qx. y-intercept of the line y = 5x+10 is.....

(A) -1/2
(B) 10
(C) 5
(D) 2

Qxi. Equation of the circle having center (3,2) and radius r=1 is given by.....

(A) $(x-3)^2+(y-2)^2=1$
(B) $(x+3)^2+(y+2)^2=1$
(C) $x^2+y^2=1$
(D) $x^2+y^2-3x-2y=1$

Qxii. Length of the tangent from $P(x_1,y_1)$ to the circle $x^2+y^2+2gx+2fy+c=0$ is.....

(A) $\sqrt{(x_1^2+y_1^2+c)}$
(B) $\sqrt{(x_1^2+y_1^2-2g-2f+c)}$
(C) $\sqrt{(x_1^2+y_1^2+2gx_1+2fy_1+c)}$
(D) None of these

Qxiii. Equation of normal to the parabola $y^2 = 4px$ at $P(x_1,y_1)$ is.....

(A) $y - y_1 = -y_1/2p\ (x - x_1)$
(B) $y - y_1 = -x_1(x - x_1)/2p$
(C) $y-y_1=2p(x-x_1)$
(D) None of these

Qxiv. $y^2/9 – x^2/16 = 1$ is the equation of .....

(A) Vertical ellipse
(B) Circle
(C) Vertical hyperbola
(D) None of these

Qxv. y=e^x +e^2x is the solution of.....

(A) y-3y=0
(B) y"-3y'+2y = 0
(C) y"-y'+y=0
(D) All of these

Qxvi. If $y = \log_{10} 9$ then y = ...

(A) 9
(B) 1/9
(C) 2
(D) -2

Qxvii. lim (x→0) (1+x)^(1/x) = ...

(A) 0 (B) 1

(C) e (D) Undefined

Qxviii. If f(x) = 16x²-7x then f'(x) = ...

(A) 32x²-7 (B) 32x²-7

(C) 32x-7 (D) 18x-7

Qxix. d/dx sec⁻¹ x = ...

(A) 1/(x√(x²-1)) (B) 1/(x√(1-x²))

(C) -1/(x√(x²-1)) (D) -1/(x√(1-x²))

Qxx. If f(x) = tanx, then f'(0) = ...

(A) 0 (B) 1

(C) –1 (D) Undefined

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 50

QQ.2. Attempt any TEN parts. All parts carry equal marks.

i. Solve the equation for x when log_b x = 3/2 log_b 4 – 2/3 log_b 8 + 2 log_b 2

ii. Find the average rate of change for y = x²-6x+5 as x increases from x=1 to x=3.

iii. Use chain rule to find dy/dx when x = 3at²+2, y=6t⁴+9.

iv. Find y" when y = (4x+2)/(3x-1)

v. Calculate lim (t→0) [(sint/t)i + ((1-cost)/t)j + e^(1-t)k]

vi. Evaluate ∫(x² – 2)e^(x³-6x+4) dx

vii. Use integration by parts to evaluate ∫xSinxdx

viii. Find the equation of a line that passes through the points A(3,1) and B(-1,-3)

ix. Find the equation of a circle that is concentric to the circle x²+y²+8x-10y+33 = 0 and is touching to the x-axis.

x. Find the equation of ellipse whose center is at (-3,2), a=2, b=1 and major axis is horizontal.

xi. Find the initial value problem when dy/dx = 2xy², y(3) = -1

xii. If z=yx f(x/y) then show that x ∂z/∂x + y ∂z/∂y = 2z

xiii. Find an interval a≤x≤b at which f(a) and f(b) have opposite sign for f(x)=e^x-2-x.

Section "C"

Maximum Marks: 30

Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ.3.

a. Evaluate limit $(\sqrt{x} - 5)/(x - 5)$ as $x \to 25$

b. Find the Maclaurin's series expansion for $f(x)=\cosh x$.

QQ.4.

a. Find dy/dx when $y = \sqrt{(\ln(1 - x^2))}$

b. Find the equation of the tangent to the circle $x^2+y^2=2$ which makes an angle of 45° with x-axis.

QQ.5.

a. Find the point of intersection between the line x-y+2=0 and parabola $y^2=9x$.

b. Evaluate the definite integral ∫xe^x dx [limits illegible in source scan]

QQ.6.

a. Reduce the differential equation $dy/dx = (y + x)^2$ in separate form and then solve it.

b. Find the perpendicular distance from the line 5x+12y-16=0 to a point P(3,-1).